

You must show work to receive credit. Do not use a calculator/computer for problems with a ★. Each problem is of equal value.

Find the slope of the line passing through the given pair of points .

1. (1, 2) and (3, -4)

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - (-4)}{1 - 3} = \frac{6}{-2} = -3$$

Use the given conditions to write an equation for the line in slope-intercept form, if possible.

2. Slope = -2, passing through (13, 4).

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 4 &= -2(x - 13) \\ y - 4 &= -2x + 26 \\ y &= -2x + 30 \end{aligned}$$

3. Passing through (-3, 0) and (-6, 4).

$$m = \frac{0 - 4}{-3 - (-6)} = \frac{-4}{3} = -\frac{4}{3}$$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 0 &= -\frac{4}{3}(x - (-3)) \\ y &= -\frac{4}{3}x + 3 \end{aligned}$$

Find the equation of the least squares line.

4. The paired data below consists of the number of Florida pleasure boats (in tens of thousands) and the number of yearly watercraft-related manatee deaths from 1995-2000.

Hours (x)	68	67	70	73	81	84
Score (y)	53	35	49	60	67	78

TI-83/Step 1: STAT>EDIT enter the "Hours" into L1 and "Score" into L2.

Step 2: STAT>CALC>LinReg or STAT>TESTS>LinRegTTest>Calculate

$$a \approx 1.9415 \text{ and } b \approx -86.3495 \Rightarrow y = 1.9415x - 86.35$$

Solve the system of two equations in two unknowns.

5. $x + y = 5$
 $x - y = 10$

$$\begin{aligned} x + y &= 5 \\ x - y &= 10 \Rightarrow 2x + 0 = 15 \Rightarrow 2x = 15 \Rightarrow x = \frac{15}{2} \end{aligned}$$

$$\text{Substituting } x: \left(\frac{15}{2}\right) + y = 5 \Rightarrow y = 5 - \frac{15}{2} \Rightarrow y = -\frac{5}{2}$$

$$\text{Solution: } \left(\frac{15}{2}, -\frac{5}{2}\right)$$

6. $x + y = 2$
 $2x + 2y = 5$

$$\begin{aligned} x + y &= 2 \\ 2x + 2y &= 5 \Rightarrow 2x + 2y = 4 \Rightarrow 2x + 2y = 4 \Rightarrow 0 = 1 \Rightarrow \text{No solution exists} \end{aligned}$$

Solve the problem.

7. There are 1000 people at a basketball game. The admission prices was \$20 for non-students and \$10 for students. The admission receipts were \$17,500. How many students and non-students attended the game?

$$\begin{aligned}
20x + 20y &= 20000 & x + y &= 1000 & \Rightarrow & 0 + 10y = 2500 & \Rightarrow & 10y = 2500 \Rightarrow y = 250 \\
20x + 10y &= 17500 & & & \Rightarrow & 20x + 10y = 17500 & & \\
10y &= 17500 & & & & & & \\
y &= 250 \Rightarrow x + (250) = 1000 \Rightarrow x = 750.
\end{aligned}$$

So there were 750 non-students and 250 students at the game.

8. Two pounds of rice and three pounds of potatoes cost a total of \$8, and three pounds of rice and one pound of potatoes cost a total of \$5. What is the price per pound for rice and the price per pound of potatoes?

$$\begin{aligned}
2x + 3y &= 8 & 2x + 3y &= 8 & \Rightarrow & 2x + 3y = 8 & \Rightarrow & x = 1 \Rightarrow 3(1) + y = 5 \Rightarrow y = 2 \\
3x + 1y &= 5 & \Rightarrow & 9x + 3y = 15 & \Rightarrow & 7x + 0y = 7 & \Rightarrow & x = 1
\end{aligned}$$

So rice costs \$1 per pound and potatoes cost \$2 per pound.

9. A car manufacturer has decided to come out with a new type of car. The fixed cost for the production of this new car is \$250,000 and the variable costs are \$15,000 per car. The company expects to sell the cars for \$20,000 each. Formulate a function $C(x)$ for the total cost of producing x new cars and a function $R(x)$ for the total revenue generated by the sales of x cars.

$$C(x) = 15000x + 250000.$$

$$R(x) = 20000x$$

10. After 5 years as an actuary, Sue's salary was \$100,000. After 10 years her salary had risen to \$200,000. Let y represent her salary after x years on the job. Assuming that a change in her salary over time can be approximated by a straight line, give an equation for this line in the form $y = mx + b$.

Years and salary give us two points: (5, 100000) and (10, 200000).

$$m = \frac{200000 - 100000}{10 - 5} = \frac{100000}{5} = 20000.$$

$$\begin{aligned}
y - 100000 &= 20000(x - 5) \\
y - 100000 &= 20000x - 100000 \\
y &= 20000x
\end{aligned}$$

11. A package delivery company delivers packages at a cost of \$5.00 per package. The fixed cost to run the delivery truck is \$450 per day. If the company charges \$7.00 per package, how many packages must be delivered to make a profit of \$500?

$$\begin{aligned}
P(x) &= R(x) - C(x) \\
&= [7x] - [5x + 450] \\
&= 7x - 5x - 450 \\
&= 2x - 450
\end{aligned}$$

need to solve $P(x) = 500$,

$$\begin{aligned}
500 &= 2x - 450 \\
950 &= 2x \\
475 &= x
\end{aligned}$$

so 475 packages.

12. A package delivery company delivers packages at a cost of \$5 per package. The fixed cost to run the delivery truck is \$450 per day. If the company charges \$7 per package, how many packages must be delivered to break even?

The profit function is the same as above. So we need to solve $P(x) = 0$.

$$\begin{aligned} 0 &= 2x - 450 \\ 450 &= 2x \\ 225 &= x \end{aligned}$$

so 225 packages.

13. Let the supply and demand functions for gasoline be given by

$$p = S(q) = \frac{4}{5}q \text{ and } p = D(q) = 300 - \frac{2}{5}q$$

where p is the price in dollars and q is the number of 42-gallon barrels. Find the equilibrium quantity and the equilibrium price.

$$\begin{aligned} S(q) &= D(q) \\ \frac{4}{5}q &= 300 - \frac{2}{5}q \\ \frac{6}{5}q &= 300 \\ q &= \frac{5}{6}300 = 250 \end{aligned}$$

Equilibrium quantity: $q = 250$

$$\frac{4}{5}(250) = 200$$

Equilibrium price: $p = \$200$

The sizes of two matrices A and B are given. Find the sizes of the product AB and the product BA , whenever these products exist.

14. A is 2×5 , and B is 5×3 .

a. AB :

$$2 \times 3$$

b. BA :

Does not exist

Find the matrix if possible.

$$15. \begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3(1) + 0(0) & 3(0) + 0(1) & 3(3) + 0(0) \\ -2(1) + 1(0) & -2(0) + 1(1) & -2(3) + 1(0) \\ 0(1) + 3(0) & 0(0) + 3(1) & 0(3) + 3(0) \end{bmatrix} = \begin{bmatrix} 3 & 0 & 9 \\ -2 & 1 & -6 \\ 0 & 3 & 0 \end{bmatrix}$$

$$16. \begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

a. Not possible. You can not multiply a (3×2) and (3×2) .

$$17. \begin{bmatrix} a & b & 1 \\ 0 & c & 0 \\ 0 & 1 & d \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ where } a, b, c, d \text{ are real numbers.}$$

$$\begin{bmatrix} a & b & 1 \\ 0 & c & 0 \\ 0 & 1 & d \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Solve the problem.

18. Let $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ Find $-2A$.

$$-2A = \begin{bmatrix} -1(-2) & 3(-2) \\ 2(-2) & 0(-2) \end{bmatrix} = \begin{bmatrix} 2 & -6 \\ -4 & -0 \end{bmatrix}$$

19. Let $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & -1 \end{bmatrix}$. Find $A + B$.

$$A + B = \begin{bmatrix} -1+0 & 3+2 \\ 2+5 & 0-1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 7 & -1 \end{bmatrix}$$

20. Let $A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 5 & -1 \end{bmatrix}$. Find $2A + B$.

$$2A + B = \begin{bmatrix} -2 & 6 \\ 4 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} -2+0 & 6+2 \\ 4+5 & 0-1 \end{bmatrix} = \begin{bmatrix} -2 & 8 \\ 9 & -1 \end{bmatrix}$$

Bonus

★ Find the inverse (A^{-1}) of $A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$ without using a calculator/computer.

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 2 & -2 & 1 \end{array} \right] \quad R2 - 2R1 \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1/2 \end{array} \right] \quad 1/2R2$$

$$\text{So } A^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & -1/2 \end{bmatrix}$$